

DETAILED DESCRIPTION OF THE INVENTION — ANTENNA DESIGN, RF PROPAGATION, AND EXTERNALIZED COMMUNICATION MODULE VARIANTS

The intraluminal physiologic monitoring device may incorporate wireless communication systems configured to transmit physiologic data to companion applications, remote platforms, or associated sensing nodes. Communication architectures may be optimized to address signal attenuation challenges associated with operation within biologic environments.

In various embodiments, wireless communication modules may include Bluetooth® Low Energy (BLE), Wi-Fi, ultra-wideband (UWB), near-field communication (NFC), proprietary low-power radio protocols, or future wireless standards capable of supporting secure physiologic data exchange.

Antenna structures may be integrated within the insertable device body, embedded along flexible conductive traces, or positioned within external pull tab assemblies extending outside the body. External antenna placement may improve signal propagation by reducing attenuation caused by surrounding tissue.

Certain embodiments may distribute communication components across internal and external regions of the device. Hybrid communication architectures may allow internal sensing modules to relay data to external antenna modules through wired interconnects, inductive coupling, optical signaling, or short-range wireless intra-device links.

Externalized communication modules may house antennas, radio-frequency amplifiers, wireless chipsets, or processing electronics within aesthetically integrated pull tab structures. Such configurations may enhance transmission reliability while minimizing internal device complexity.

Adaptive communication systems may dynamically adjust transmission power, frequency selection, data compression, or transmission scheduling based on signal quality measurements or energy availability.

Communication security frameworks may employ encryption, secure pairing protocols, rotating identifiers, or authenticated communication channels to protect sensitive physiologic information during wireless transmission.

Multi-device communication architectures may enable synchronization between multiple physiologic sensing nodes, companion wearables, or remote monitoring systems forming coordinated sensing networks.

Future embodiments may incorporate mesh networking, relay devices, body-area networks, or environmental gateways allowing physiologic data transmission through distributed communication infrastructure.

The antenna and wireless communication architectures described herein are illustrative rather than limiting. Any method enabling wireless transmission of intraluminal physiologic monitoring data is considered within the scope of the present disclosure.